DG-MCTS: Dual-Guidance Monte Carlo Tree Search for Adaptive Emotional Support Dialogue Planning

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Abstract

With the increasing popularity of online platforms, the Web has become a primary venue for support and emotional expression, prompting conversational agents to provide both factual and emotional assistance. Emotional Support Conversation (ESC) systems aim to alleviate users' emotional distress while simultaneously helping them identify and overcome the underlying problems. Existing approaches focus largely on modeling emotions and producing empathetic responses, frequently overlooking problem-solving. Moreover, LLMs exhibit inherent biases, often favoring certain strategies—such as prioritizing emotional support over problem-solving—which hinder balanced pursuit of dialogue objectives. Consequently, existing approaches without long-term planning struggle to coordinate dual objectives across multiple turns, limiting their ability to provide comprehensive support. To address these challenges, we propose Dual-Guidance Monte Carlo Tree Search (DG-MCTS) for enhancing ESC systems. By modeling dialogue generation as a tree search guided by emotional support and problem-solving signals, DG-MCTS achieves explicit decoupling and dynamic balancing of these dual objectives. To maintain strategic coherence across multiple turns while remaining responsive to evolving user states, we employ a hierarchical strategy mechanism, where coarse-grained orientation selection guides the overall dialogue toward emotional support or problem-solving, and fine-grained strategy adaptation addresses immediate conversational needs within the chosen orientation. Furthermore, the cumulative scoring continuously tracks historical attention to both objectives along dialogue paths, dynamically shifting focus toward insufficiently explored goals. This ensures balanced pursuit of emotional support and problem-solving while enhancing the effectiveness of strategy execution. We evaluate DG-MCTS on widely used ESC datasets and in online deployment, and experimental results demonstrate that our method achieves state-of-the-art performance.

CCS Concepts

• **Computing methodologies → Discourse, dialogue and pragmatics**.

Keywords

Emotional Support Conversation, Dialogue Systems, Monte Carlo Tree Search

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1 Introduction

With the rapid development of online platforms and social networks, the Web has become a central medium for people to seek information, share experiences, and express emotions. Consequently, conversational agents are increasingly expected not only to provide factual assistance but also to offer emotional support, giving rise to the task of Emotional Support Conversation (ESC), which has attracted growing attention in the research community [21]. An ESC system is designed to reduce users' emotional distress and help them identify and overcome problems through conversations. Integrating such emotional support capabilities into dialogue systems renders them highly beneficial in a wide range of application scenarios and significantly enhances user experience [6, 11, 17].

Existing methods primarily adopt emotion-centered approaches, which focus on explicitly modeling emotions and generating empathetic responses to address users' needs. Some methods infer users' emotions by incorporating external knowledge [12, 15, 28, 32]. MISC [28] innovatively integrates COMET[2], harnessing external commonsense knowledge to enhance the model's adeptness in reasoning about emotions. Hao and Kong [15] propose dynamic knowledge filtering to select contextually relevant commonsense to improve emotion reasoning. Other methods design specific modules

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for emotion detection [8, 36]. Cheng et al. [8] employ a specialized detector to obtain the cause of the emotion. Zhao et al. [36] explore state transitions between each conversation turn. Despite their success in generating empathetic responses, existing emotion-centered approaches primarily focus on emotional modeling rather than problem-solving. To truly address users' emotional needs, it is crucial to identify and resolve the underlying problems that cause these emotions, rather than solely providing empathetic responses based on emotional states.

Recent efforts have begun to leverage LLMs to address both emotion support and problem-solving simultaneously[6, 9, 31]. Chen et al. [6] use prompting to control LLM generation for mixed-initiative dialogue that combines emotional support with goal-directed conversation management. COOPER[9] coordinates multiple specialized agents to jointly address different aspects of complex dialogue goals. However, LLMs' preference biases and strong predisposition towards certain strategies often cause them to struggle with balancing multiple objectives, undermining overall dialogue outcomes[1, 18]. These limitations necessitate introducing long-term planning capabilities to LLMs. ESC requires resolving both emotional distress and underlying problems by conversation's end, demanding strategic planning to coordinate dual objectives across dialogue trajectories rather than responding reactively to isolated turns.

Monte Carlo Tree Search (MCTS) presents a compelling solution to these planning challenges by naturally incorporating long-term outcome evaluation into the decision-making [3, 26, 27]. MCTS excels at planning by systematically exploring possible future trajectories through iterative tree expansion and evaluation, allowing systems to look ahead multiple steps and select optimal actions for long-term goals. This capability has been successfully applied to various multi-step reasoning tasks, such as mathematical reasoning [5, 35] and game playing [13], demonstrating its potential for dialogue systems where current decisions should consider their impact on future conversational trajectories. Consequently, some studies combine MCTS with LLMs to enhance their lookahead planning capabilities in dialogue tasks. However, these existing works typically model dialogue planning as a single-objective optimization problem, such as focusing on persuasion [16, 33] or recommendation [10], which suits their respective tasks. In contrast, ESC tasks inherently require addressing both users' immediate emotional distress and the underlying problems causing these emotions, creating a fundamental challenge of strategically balancing dual objectives throughout the dialogue trajectory.

To enable LLMs to perform long-term dialogue planning while dynamically balancing emotional support and problem-solving, we propose a dual-guidance Monte Carlo Tree Search (DG-MCTS) framework for ESC tasks. Our approach models the dialogue generation process as a tree search guided by dual value signals: emotional support value and problem-solving value. DG-MCTS explicitly decouples these dual objectives to enable proactive trajectory planning. This allows the system to dynamically balance dual objectives rather than reacting passively to recent turns. To maintain strategic coherence across multiple turns while remaining responsive to evolving user states, we employ a hierarchical strategy mechanism that operates at two levels. Coarse-grained orientation selection guides the overall dialogue direction toward either emotional support or problem-solving, while fine-grained strategy adaptation responds to immediate conversational needs within the chosen orientation. To achieve dynamic objective balancing, we introduce a cumulative scoring system that tracks historical attention to both objectives along dialogue paths. The framework automatically shifts focus toward underexplored objectives, preventing over-concentration on single goals. The dual-value propagation mechanism maintains independent Q-values for emotional support and problem-solving, enabling separate optimization of both objectives without mutual interference. This design allows the system to simultaneously pursue both goals while ensuring they remain aligned rather than conflicting. Furthermore, to continuously improve strategy effectiveness, the system employs a feedback mechanism that refines strategy selection based on evaluation outcomes. Experiments on widely used ESC datasets, as well as in online deployment, show that our method surpasses strong baselines. Our main contributions are as follows:

- We propose a novel dual-guidance Monte Carlo Tree Search framework that explicitly models and balances emotional support and problem-solving objectives in ESC, enabling proactive and adaptive dialogue planning rather than purely reactive responses.
- We design a dual-value hierarchical planning mechanism that integrates coarse-grained orientation selection with fine-grained sub-strategy adjustment, coupled with independent value estimation and backpropagation for each objective, thus ensuring balanced and context-aware strategy transitions.
- We achieve state-of-the-art performance on widely used ESC datasets and in online deployment, demonstrating the effectiveness of our approach.

2 Related Work

2.1 Emotional Support Conversation

Emotional Support Conversation is a task that aims to help users alleviate emotional distress and address their problems through dialogue [21]. Existing methods primarily rely on recent dialogue context, especially the current turn, augmenting responses with auxiliary information and specialized components. Some methods [12, 20, 28, 31, 32] leverage external commonsense knowledge to infer the user's immediate mental state from the current dialogue context, often relying on COMET [2], a pre-trained generative model for commonsense reasoning. Additionally, other methods integrate personalized character information to strengthen role modeling, ensuring that responses are empathetically tailored and consistent with the user's background [7, 15]. Another set of methods designs dedicated emotion recognition modules, ensuring that the system can accurately detect and interpret shifts in the user's emotional state during the conversation [8, 29, 36]. However, these methods are essentially reactive, focusing on the current turn without long-term dialogue planning, making it difficult to dynamically adjust strategies and balance emotional support with problem-solving over multiple turns.

2.2 MCTS for Dialogue Planning

Monte Carlo Tree Search has emerged as a powerful technique for decision-making and planning in complex domains, enabling models to perform structured exploration and long-term optimization for more strategic choices [5, 34, 35]. Building on this success, recent works have begun to introduce MCTS into the field of dialogue planning. These approaches typically model dialogue planning as a single-objective optimization problem. For instance, Yu et al. [33] introduced GDP-ZERO, an open-loop MCTS framework that enables goal-oriented dialogue planning without any model training. He et al. [16] proposed a framework that combines a fast, instinctive policy model with a deliberative MCTS mechanism for dialogue planning. Dao et al. [10] proposed Experiential Policy Learning, a framework that leverages past dialogue experiences to improve target-driven recommendation. Despite these advances, prior methods mainly focus on single-objective planning and rarely address multi-objective, adaptive control in dialogue. In contrast, ESC requires balancing multiple objectives—providing emotional support while guiding users toward problem-solving over multiple turns. To address this gap, we design a dual-guidance MCTS framework that jointly models emotional support and problem-solving, with hierarchical strategy control and dual-value estimation, enabling dynamic, context-aware, and balanced multi-turn dialogue planning.

3 Method

3.1 Overview

In the task of Emotional Support Conversation, given a dialogue context $C = [u_1, u_2, \ldots, u_{n-1}]$, where each $u_i = (u_{ui}, u_{si})$ contains a user utterance and system reply, the objective is to generate the forthcoming response u_{sn} based on the current utterance u_{un} and the preceding $n-1$ turns. The core goals of this task are twofold: providing emotional support to alleviate the user's distress and offering problem-solving guidance to help the user address their issues. Our DG-MCTS framework consists of two training stages. The first one is DG-MCTS sampling for self-training, which uses MCTS to generate balanced dialogue trajectories and fine-tune the model for dual-objective coordination. The second stage is task-specific supervised fine-tuning to enhance context-aware response generation.

3.2 DG-MCTS Sampling for Self-Training

Our DG-MCTS framework samples high-quality training data through four stages: selection, expansion, evaluation, and backpropagation. The resulting balanced dialogue trajectories are used to train our generation model, enabling dual-objective coordination without intensive search during inference. The relevant prompt templates are provided in the Figure 8 and 9.

The root node s_0 is initialized with the first k dialogue turns. Each node $s \in \mathcal{T}$ represents a single turn of interaction between the user and the system. The search tree is expanded via MCTS to efficiently explore and evaluate potential future dialogue paths, guided by both emotion-focused and problem-focused strategies. The search proceeds until a predefined number of turns is reached or the dialogue ends.

3.2.1 Selection. The goal of the selection phase is to identify a node $s \in \mathcal{T}$ from the partial search tree constructed so far for further exploration.

To this end, the method employs a Dual-Guided UCT, which combines guidance from emotional support and problem solving with a consistency score. At time step t, the node s_t maintains two scalar values: $Q_e(s_t)$ denotes the emotional support value and $Q_q(s_t)$ denotes the problem-solving value. A consistency score $Q_{\text{align}}(s_t)$ is also introduced to measure the alignment between the two guidance signals. The overall selection score combines these components as follows:

$$U(s_t) = Q_e(s_t) + Q_q(s_t) + c \cdot \frac{\sqrt{N(s_{t-1})}}{1 + N(s_t)} + \alpha \cdot Q_{\text{align}}(s_t) \quad (1)$$

The consistency score $Q_{\text{align}}(s_t)$ is calculated as:

$$Q_{\text{align}}(s_t) = \frac{\min(Q_e(s_t), Q_q(s_t))}{1 + |Q_e(s_t) - Q_q(s_t)|} \quad (2)$$

Here, $N(s_{t-1})$ and $N(s_t)$ refer to the visit counts of the parent node and the current node. The constant c is the exploration coefficient, and α is the consistency weighting factor that balances the influence of alignment in the overall selection score.

3.2.2 Expansion. Following the selection of an expandable node via Dual-Guided UCT, the expansion phase generates multiple candidate child nodes to explore a wider spectrum of conversational paths. A key innovation in our approach is the dynamic adjustment of the expansion direction based on identifying underexplored guidance signals—either emotional support or problem-solving—along the current dialogue path. To mitigate the risk of over-focusing on a single guidance signal, we employ a hierarchical mechanism that is specifically designed to dynamically balance between emotional support and problem-solving objectives.

At the coarse-grained level, guidance orientation adaptively selects between emotional support and problem-solving as the expansion direction. At the fine-grained level, specific sub-strategies are then selected to adapt to the user's evolving needs within the chosen orientation. Each coarse-grained guidance signal maps to a set of fine-grained sub-strategies, forming a one-to-many relationship.

To enable adaptive orientation selection, our method employs two types of value measures. Q-values assess immediate dual-objective quality at individual nodes, while cumulative scores z-values track historical objective attention across the dialogue path. The z-values aggregate Q-values from previous nodes, revealing which objective has been neglected over the conversation history. This guides expansion toward underexplored objectives. The cumulative scores are formulated as:

$$z_t^e = Q_e(s_t) + \sum_{j \in \text{Path}(s_t)} \lambda^j Q_e(s_j) \quad (3)$$

$$z_t^q = Q_q(s_t) + \sum_{j \in \text{Path}(s_t)} \lambda^j Q_q(s_j) \quad (4)$$

where $\text{Path}(s_t)$ denotes the sequence of nodes from the root s_0 to node s_{t-1}, and $\lambda \in (0, 1)$ is a decay factor that discounts distant historical turns.

Thus, a coarse-grained orientation distribution $[P_t^e, P_t^q]$ is computed based on the cumulative scores z_t^e and z_t^q. This distribution

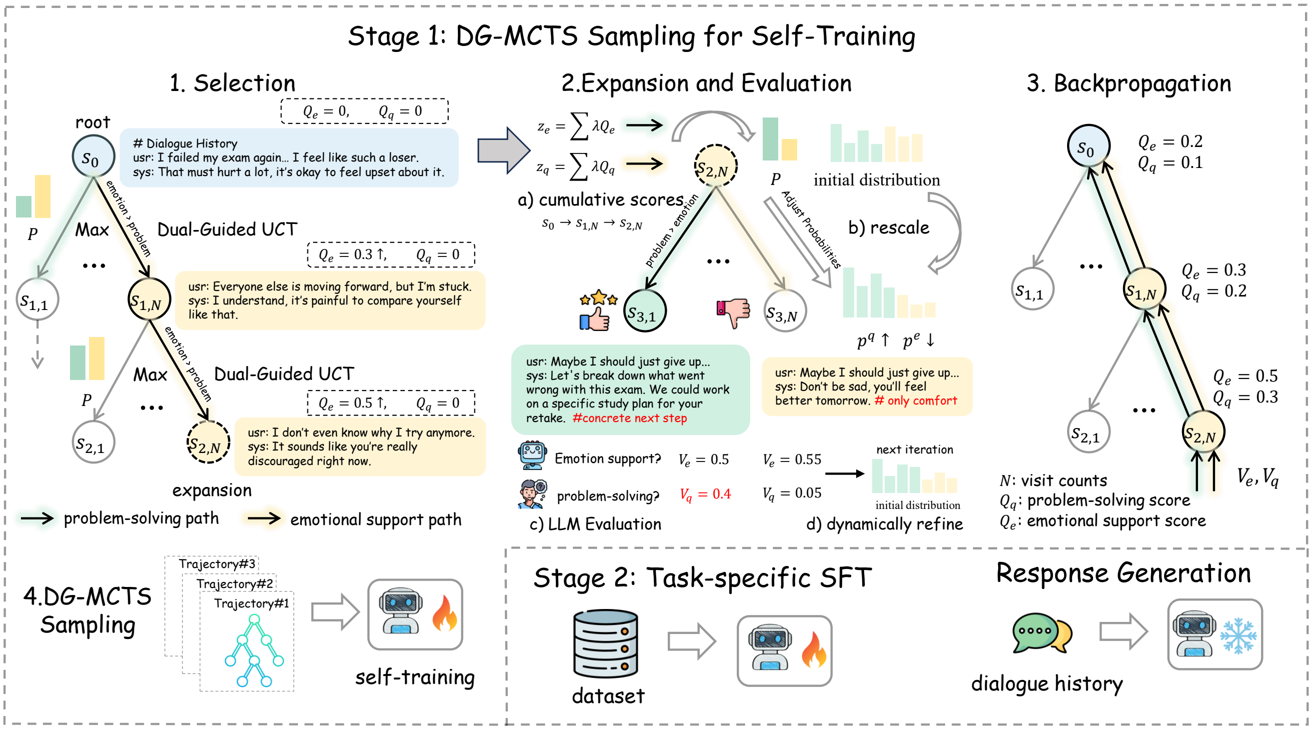

Figure 1: Our DG-MCTS framework consists of two training stages. The first one is DG-MCTS sampling for self-training, which uses MCTS to generate balanced dialogue trajectories and fine-tune the model for dual-objective coordination. The second stage is task-specific supervised fine-tuning to enhance context-aware response generation.

indicates the relative lack of each guidance type and serves as a higher-level signal for adjusting the expansion direction. The orientation probabilities $[P_t^e, P_t^q]$ are obtained by applying softmax to cumulative scores $[z_t^e, z_t^q]$ and then inverting the result, so that higher values reflect a stronger preference toward the corresponding objective.

To align fine-grained strategy selection with the high-level guidance, the initial sub-strategy distribution is modulated based on the coarse-grained orientation signal. We first derive an initial distribution p_{a_t} estimated from dialogue transition patterns in the training set, which captures the likelihood of moving from the current strategy a_t at time step t to each candidate sub-strategy. Formally, let $p_{a_t} = [p^e_{a_t \rightarrow i}, ..., p^q_{a_t \rightarrow i}, ...]$ be the initial probabilities at time step t over emotion-focused and problem-focused sub-strategies.

The fine-grained sub-strategy distribution is rescaled based on the coarse-grained orientation probabilities $[P_t^e, P_t^q]$ to dynamically adjust strategy preferences under high-level guidance. This modulation enhances the relative importance of strategies in underexplored directions by increasing the transition probabilities toward insufficiently explored areas while decreasing those toward already well-explored ones, thereby enhancing the likelihood of selecting actions aligned with the dominant guidance signal. The rescaled strategy distributions are computed as:

$$\tilde{p}^g_{a_t \rightarrow i} = \frac{p^g_{a_t \rightarrow i} / (1 - P_t^g)}{\sum_j^{N_g} p^g_{a_t \rightarrow j} / (1 - P_t^g)}, \quad g \in \{e, q\} \tag{5}$$

Here, N_g denotes the number of emotion-focused or problem-focused sub-strategies, respectively. Finally, $\tilde{p}^g_{a_t \rightarrow i}$ is normalized to form a valid fine-grained strategy distribution for guiding sub-strategy selection in the expansion phase at step $t+1$.

3.2.3 Evaluation. In the evaluation phase, our method assesses the effectiveness of each candidate response based on its dual communicative goals: problem-solving and emotional support. To this end, we leverage the LLM to evaluate the current dialogue context and generate two scalar scores: the problem-solving score V_q and the emotional support score V_e. Specifically, scoring is performed using a custom evaluation prompt, in which the LLM is presented with exemplar responses corresponding to the same user request annotated with different scores, and is then instructed to assess the candidate response within the context of the current dialogue. These scores quantify the degree to which a response satisfies user requests and alleviates emotional distress, respectively. Details of the LLM-based evaluation prompt templates are provided in the Figure 10 and Figure 11.

Beyond serving as supervision signals for backpropagation, the evaluation results are also used to dynamically refine the initial sub-strategy distribution p_{a_t} introduced in Section 3.2.2. By suppressing low-performing transitions and reinforcing high-performing ones, this refinement enables the transition probabilities to better capture strategy utility and thus provide more reliable guidance for subsequent node expansion. Specifically, we adjust the strategy transition probabilities based on evaluation feedback aggregated from the expanded child nodes. Let $p_{a_t \to i}$ denote the transition probability from strategy a_t to strategy i. The update process for the score $p_{a_t \to i}$ is as follows:

$$p_{a_t \to i} = \text{Normalize}(p_{a_t \to i} + \eta \cdot \frac{\sum V_e(a_t, i) + \sum V_q(a_t, i)}{2N(a_t, i)}) \quad (6)$$

where $V_q(a_t, i)$ and $V_e(a_t, i)$ are the score V_q and the emotional support score V_e for transitions from strategy a_t to strategy i, $N(a_t, i)$ is the number of transitions from strategy a_t to strategy i, and η is the weighting factor for updating.

3.2.4 Backpropagation. After completing a simulation along a search path, the resulting evaluation signals $V_e(s_{t+1})$ and $V_q(s_{t+1})$ are propagated backward through the tree $\mathcal{T}$. During this phase, the values associated with each node on the path are updated to reflect its contribution to overall dialogue quality. To ensure independent handling of emotional support and problem-solving signals, our method applies separate value updates for each guidance type. This design avoids interference between the dual objectives and preserves the interpretability of Q-values. Formally, for each node s_i along the backpropagation path, the update rules are defined as:

$$N(s_i) \leftarrow N(s_i) + 1, \quad i = 0, ...t + 1 \quad (7)$$

$$Q_e(s_i) \leftarrow Q_e(s_i) + \frac{1}{N(s_i)}\left(V_e(s_{t+1}) - Q_e(s_i)\right) \quad (8)$$

$$Q_q(s_i) \leftarrow Q_q(s_i) + \frac{1}{N(s_i)}\left(V_q(s_{t+1}) - Q_q(s_i)\right) \quad (9)$$

Here, $N(s_i)$ denotes the visit count of node s_i, while $Q_e(s_i)$ and $Q_q(s_i)$ represent its emotional support and problem-solving value estimates, respectively. $V_e(s_{t+1})$ and $V_q(s_{t+1})$ are the corresponding target values of the node s_{t+1}, corresponding to the state at time step $t+1$. Each update incrementally refines the node's utility under the dual-guidance framework, enabling more effective future selection.

3.2.5 Self-training. The self-training stage is designed to enable the backbone model to internalize the benefits of DG-MCTS, allowing it to generate responses that effectively balance emotional support and problem-solving without relying on intensive search at inference time. Thus, we select the high-quality dialogue trajectories generated during the prior DG-MCTS process and employ them as training data $\mathcal{D}_{DG-MCTS} = \left\{(q_i^{mcts}, \text{concat}(a_i^{mcts}, r_i^{mcts}))\right\}_{i=1}^{N}$. The backbone model is trained to jointly generate the dialogue strategy a^{mcts} and its corresponding response r^{mcts} for each user query q^{mcts}. Specifically, the extracted pairs are reformulated into supervised fine-tuning (SFT) samples by incorporating role-specific prompts C_r to adapt the model to the task scenario, along with explicit strategy-selection instructions C_a that guide the model to generate responses appropriate to the given user query. The self-training objective is formalized as follows:

$$\max \mathbb{E}_{(q^{mcts}, z^{mcts}) \sim \mathcal{D}_{\text{DG-MCTS}}} \left[\log p(z^{mcts} \mid q^{mcts}, C_r, C_a)\right] \quad (10)$$

where $z^{mcts} = [a^{mcts} \,\|\, r^{mcts}]$ concatenates the strategy and the response.

3.3 Task-specific SFT and Response Generation

3.3.1 Task-specific SFT. Through the self-training stage using MCTS-sampled data, we endow the model with long-term dialogue planning capabilities across multiple objectives, as the sampled trajectories are generated to jointly optimize both emotional support and problem-solving. To enhance its context-aware response generation capabilities, we further optimize the backbone LLMs via task-specific supervised fine-tuning (SFT). Specifically, prompt templates C_e are carefully designed as hints to guide the model in comprehending the meaning and purpose of various dialogue strategies. By incorporating dialogue history, the model can anticipate subsequent conversational turns and generate contextually appropriate responses. Standard negative log-likelihood (NLL) loss is adopted for the ground truth response during the finetuning process:

$$\mathcal{L}_{\text{task}} = -\frac{1}{M}\sum_{j=1}^{M}\sum_{t=1}^{|y_j|} \log p\left(y_{j,t} \mid y_{j,<t},\, h_j,\, C_e\right), \quad (11)$$

where h_j denotes the dialogue history, and y_j is the ground truth response.

3.3.2 Response Generation. After the two-stage training, the fine-tuned LLMs are employed to anticipate the imminent dialogue future based on dialogue history. For response generation, we append role-alignment and response-guidance instructions to the dialogue context to compose the input of the model, guiding it to generate coherent and appropriate responses. The detailed prompt templates are provided in the Figure 12.

4 Experiments

4.1 Setting

Datasets. We conduct experiments on two benchmark datasets: Emotional Support Conversation (ESConv) [21] and EMPATHEticDIALOGUES (ED) [24]. ESConv comprises approximately 1,053 multi-turn dialogues between a help seeker facing emotional distress and a professional supporter. ED is a larger dataset containing 25,000 multi-turn empathetic dialogues between a speaker and a listener.

Implementation. For the backbone models, we employed LLaMA-3.1-8B-instruct and Qwen-2.5-7B-instruct. All training was performed using the LLaMAFactory platform. Further implementation details are provided in Appendix D.

4.2 Baseline Methods

We compare DG-MCTS with several state-of-the-art prompt-based and SFT-based method: **M-Cue CoT** [30], **COMET** [2], **DOCTOR** [4], **DIALeCT** [25], **GDP-MCTS** [33], **ECoT** [19], **Strategy Planner** [18], **EmoStage** [22], **Sibyl** [31], **LLaMA3.1** [14], **Qwen2.5** [23]. The detail of the above baseline methods is specified in Appendix A.

4.3 Automatic Evaluation

For automatic evaluation, we employed a comprehensive set of metrics to evaluate the responses. Specifically, ROUGE-L measures n-gram overlap and recall between generated and reference responses; BLEU-3 and BLEU-4 evaluate the precision of n-gram

Model Class	Model	ROUGE-L	BLEU-3	BLEU-4	Dist-1	MET.	Ext.	CIDEr
Prompt-based Methods	llama3.1-8b(2024)	13.52	2.38	1.56	7.32	10.59	44.70	4.11
	qwen2.5-7b(2025)	14.91	2.71	1.80	7.54	10.02	44.94	7.24
	qwen2.5-72b(2025)	14.30	2.50	1.62	4.37	**11.02**	43.98	4.37
	gpt4.0(2024)	14.86	2.01	0.93	6.43	8.50	41.90	4.22
	+comet(2019)	14.87	1.99	0.91	5.98	9.44	42.98	4.14
	+DOCTOR(2023)	13.98	1.72	0.79	6.36	8.73	40.93	3.39
	+DIALeCT(2022)	14.97	1.82	0.81	6.42	9.10	42.56	4.15
	+M-cue CoT(2023)	14.79	1.89	0.92	6.32	9.27	41.73	3.98
	+Sibyl(2025)	15.20	2.21	1.10	6.52	9.65	41.90	4.95
SFT-based Methods	llama3.1-8b(2024)	15.62	2.92	1.41	6.24	9.12	44.71	8.76
	+comet(2019)	15.58	2.78	1.35	6.22	9.04	45.00	9.34
	+DOCTOR(2023)	15.78	2.83	1.42	6.68	8.24	45.14	9.84
	+DIALeCT(2022)	16.02	2.79	1.29	6.35	8.22	44.86	10.44
	+GDP-MCTS(2023)	17.15	3.13	2.20	7.55	7.10	49.16	13.66
	+ECOT(2024)	14.47	2.71	1.81	4.88	7.89	47.85	6.49
	+EmoStage(2025)	17.46	3.61	2.48	6.99	8.40	49.50	13.80
	+Strategy Planner(2024)	16.87	3.37	2.36	7.50	8.13	49.74	14.08
	+Sibyl(2025)	16.23	3.04	1.52	6.84	8.53	45.86	10.92
	DG-MCTS+qwen2.5(Ours)	18.10	3.82	2.66	7.60	9.23	50.04	15.76
	DG-MCTS+llama3.1(Ours)	**19.23**	**4.20**	**3.03**	**8.10**	8.73	**50.41**	**18.89**

Table 1: Automatic Evaluation results on ESConv dataset. The best results of each metric are bolded, and the second best are underlined.

Model Class	Model	ROUGE-L	BLEU-3	BLEU-4	Dist-1	MET.	Ext.	CIDEr
Prompt-based Methods	llama3.1-8b(2024)	11.08	1.80	1.20	5.86	8.97	44.31	1.91
	qwen2.5-7b(2025)	11.95	2.06	1.39	6.05	8.23	45.89	5.02
	qwen2.5-72b(2025)	12.11	2.04	1.37	5.98	8.78	44.03	3.98
	gpt4.0(2024)	14.01	2.28	1.21	9.14	**10.66**	46.66	7.90
	+M-cue CoT(2023)	12.64	1.67	0.79	9.30	9.29	44.66	5.72
	+COMET(2019)	14.89	2.43	1.34	9.36	9.13	45.69	7.54
	+DOCTOR(2023)	15.65	2.66	1.49	9.68	9.30	46.24	8.43
	+DIALeCT(2022)	16.23	2.64	1.39	8.98	9.46	47.47	10.48
	+Sibyl(2025)	17.62	3.21	1.83	**9.70**	10.09	48.13	14.14
SFT-based Methods	llama3.1-8b(2024)	15.16	2.70	1.45	5.63	7.60	48.08	13.73
	+COMET(2019)	15.31	2.89	1.59	5.61	7.74	48.45	14.47
	+DOCTOR(2023)	15.16	2.85	1.58	5.62	9.15	48.25	13.60
	+DIALeCT(2022)	17.37	4.14	2.42	5.53	8.61	49.81	22.21
	+GDP-MCTS(2023)	18.08	3.93	2.74	5.85	10.12	50.29	16.61
	+ECOT(2024)	18.98	4.05	2.90	6.48	9.45	50.10	20.41
	+EmoStage(2025)	13.57	2.47	1.71	4.28	7.26	47.61	7.19
	+Sibyl(2025)	19.08	**5.01**	2.95	5.65	9.61	50.90	**26.93**
	DG-MCTS+qwen2.5(Ours)	19.15	3.94	2.82	5.74	9.40	51.49	20.14
	DG-MCTS+llama3.1(Ours)	**20.21**	4.69	**3.39**	6.65	10.21	**51.66**	24.85

Table 2: Automatic Evaluation results on the ED dataset. The best results of each metric are bolded, and the second best are underlined.

matches; Distinct-1 captures response diversity; CIDEr quantifies consensus with multiple references; METEOR considers both lexical matching and semantic similarity; and Extrema similarity (Ext.) measures semantic similarity based on embeddings.

Table 1 shows the results on the Emotional Support Conversation (ESC) task. Integrating DG-MCTS with the backbone models consistently outperforms all baseline methods, demonstrating the effectiveness of our approach in emotional support dialogue generation. Specifically, DG-MCTS achieves higher similarity scores (ROUGE-L, BLEU, Ext., CIDEr) and diversity scores (Distinct) compared to all baselines. Performance on METEOR remains comparable to the best-performing baselines. Compared to GDP-MCTS, which applies MCTS-based planning for dialogue modeling, DG-MCTS achieves stronger results by jointly optimizing emotional feedback and problem-solving. Notably, some baselines, such as COMET and Sibyl, leverage external knowledge sources or closed-source LLMs to acquire commonsense knowledge for predicting dialogue trajectories. In contrast, DG-MCTS achieves superior performance without incorporating any external knowledge, relying solely on dual-objective optimization. This further underscores the critical role of the dual-objective framework in guiding response generation. Moreover, applying DG-MCTS to mainstream LLMs such as LLaMA3 and Qwen2.5 further enhances performance, highlighting the generalizability of our method.

On the ED dataset, we compared DG-MCTS against relevant baselines to evaluate empathetic response generation. As shown in Table 2, DG-MCTS improves the backbone model's performance across key metrics for empathetic responses. When compared with supervised fine-tuning baselines, DG-MCTS consistently achieves the highest ROUGE and Extrema scores, and competitive BLEU and CIDEr scores, demonstrating its ability to generate high-quality responses. Against prompt-based baselines, although diversity and METEOR scores are slightly lower due to model scale limitations, DG-MCTS outperforms on other evaluation metrics, emphasizing the distinctive effectiveness of our proposed paradigm in generating empathetic responses. Nevertheless, the overall improvements are less pronounced on the ED dataset, as the dialogues are generally shorter and the response patterns more constrained, limiting DG-MCTS's ability to fully leverage its long-term planning and multi-objective optimization capabilities.

	ED			ESConv		
	Nat	Emp	Coh	Nat	Sup	Coh
llama3.1-SFT	2.512	1.849	2.635	2.332	1.376	2.214
COMET	2.464	1.747	2.646	2.368	1.944	2.465
DOCTOR	2.503	2.088	2.653	2.349	1.408	2.496
DIALeCT	2.441	1.115	2.644	2.381	1.867	2.526
Sibyl	2.568	2.396	2.774	2.387	1.958	2.599
GDP-MCTS	2.957	2.424	2.945	2.872	2.264	2.743
DG-MCTS(Ours)	**2.977**	**2.465**	**2.954**	**2.901**	**2.45**	**2.861**

Table 3: LLMs based Evaluation results on ED and ESConv dataset under Supervised Finetuning. The best results of each metric are bolded.

4.4 LLM-based Evaluation

Following [31], we employed G-Eval to assess the naturalness (Nat.) and coherence (Coh.) of responses generated by different baseline methods. For task-specific requirements, we further measured empathy (Emp.) on the ED dataset and supportiveness (Sup.) on the ESConv dataset. We compared DG-MCTS against the following baselines: COMET and Sibyl use external knowledge to generate contextually relevant responses, DOCTOR and DIALeCT perform internal reasoning to consolidate conversational information, and GDP-MCTS plans via MCTS with LLM-based components.

For evaluation, we randomly sampled 200 data from both ED and ESConv datasets. Weighted average scores were computed over the sampled data. The comparison results, reported in Table 3, indicate that DG-MCTS consistently outperforms other baselines across all metrics, highlighting the effectiveness of the dual-objective optimization framework in generating empathetic and supportive responses.

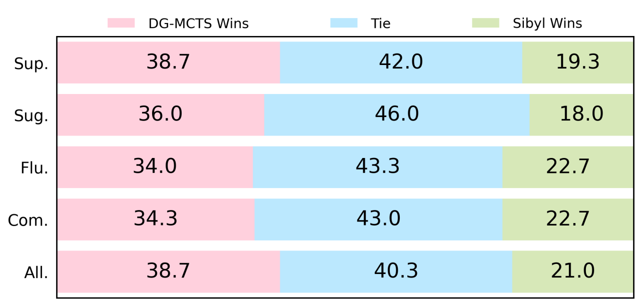

Figure 2: Human A/B test of ESConv.

4.5 Human Evaluation

Following [31], we conducted human evaluations to assess the quality of generated responses on both the EmpatheticDialogues (ED) and ESConv datasets. More details are provided in Appendix B.

As illustrated in Figure 2 and Figure 3, DG-MCTS consistently outperforms Sibyl, the current state-of-the-art (SOTA) model, across all evaluation dimensions. Notably, whereas Sibyl relies on GPT-4o to extract external commonsense knowledge to enhance response generation, DG-MCTS achieves superior human-judged performance solely through dual-objective optimization, further demonstrating the effectiveness of our approach in generating empathetic and supportive dialogue responses.

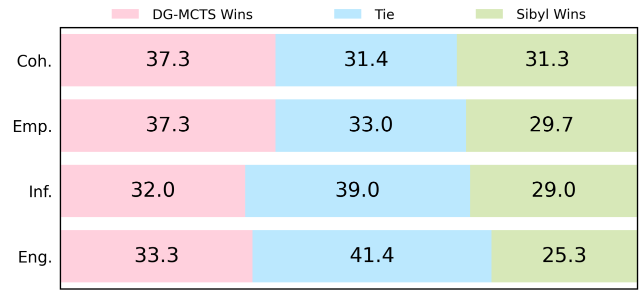

Figure 3: Human A/B test of EMPATHETICDIALOGUES.

4.6 Ablation Study

To investigate the contribution of different components within DG-MCTS, we conducted a systematic ablation study on both ED and ESConv datasets. Specifically, we considered the following two ablation variants:
1)w/o dual-grained distribution: Removing the hierarchical strategy adjustment component in the MCTS expansion stage.
2)w/o dual-guided value: Eliminating the dual value at each MCTS node, relying solely on a single general value.
3)w/ tree search: Replacing the Monte Carlo Tree Search with tree search, to evaluate the contribution of the MCTS-based mechanism.

The ablation results, shown in Tables 4 and 5, indicate that removing any component leads to a clear decline in response quality. Although certain variants may outperform the DG-MCTS model on specific metrics, their overall performance is substantially lower, highlighting the necessity of each component. Removing the dual-guided value or replacing MCTS with tree search results in a marked performance decline, confirming that dual-value-based node selection and backpropagation are essential for jointly optimizing emotional support and problem-solving objectives in dialogue generation.

4.7 Case Study

Table 7 presents two illustrative examples demonstrating that DG-MCTS generates responses that are more supportive and empathetic than those generated by LLaMA-3.1-SFT and Sibyl.

In the first case, DG-MCTS not only acknowledges the speaker's desire for social acceptance but also subtly highlights the importance of being treated with respect, striking a balance between empathy and practical guidance. In the second case, while all models recommend face-to-face communication with parents, DG-MCTS further clarifies how in-person interactions can enhance emotional expression and mutual understanding, yielding a more engaging and contextually appropriate response. These examples underscore that DG-MCTS consistently generates responses that are supportive, empathetic, and attuned to the emotional context, highlighting its ability to deliver nuanced, context-aware guidance in dialogue.

Model	ROUGE-1	ROUGE-2	ROUGE-L	BLEU-3	BLEU-4	Dist-1	MET.	Ext.	CIDEr
DG-MCTS	**22.70**	**6.26**	**19.23**	**4.20**	**3.03**	8.10	**8.73**	**50.41**	**18.89**
w/o dual-grained distribution	22.21	5.94	18.74	4.00	2.84	**8.66**	8.27	50.14	17.21
w/o dual-guided value	22.38	6.19	18.94	4.16	2.91	8.19	8.62	50.32	17.51
w/ tree search	22.31	6.06	18.86	4.08	2.87	8.17	8.46	50.18	17.25

Table 4: Ablation study on the ESConv dataset. The best results of each metric are bolded.

Model	ROUGE-1	ROUGE-2	ROUGE-L	BLEU-3	BLEU-4	Dist-1	MET.	Ext.	CIDEr
DG-MCTS	**22.23**	**7.00**	**20.21**	**4.69**	**3.39**	6.65	10.21	**51.66**	**24.85**
w/o dual-grained distribution	21.90	6.64	19.88	4.56	3.27	7.00	10.15	51.24	23.63
w/o dual-guided value	21.95	6.83	19.88	4.61	3.30	**7.12**	**10.26**	51.25	24.16
w/ tree search	21.83	6.57	19.77	4.67	3.35	6.85	10.15	51.16	24.33

Table 5: Ablation study on the ED dataset. The best results of each metric are bolded.

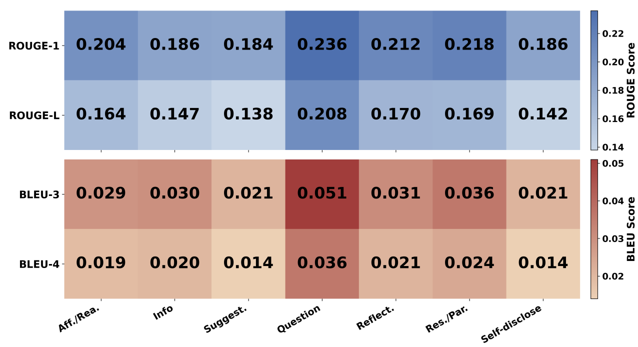

Figure 4: Performance of DG-MCTS across individual dialogue strategies.

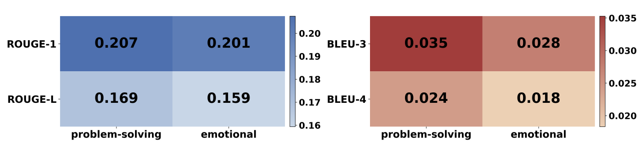

Figure 5: Categorized performance of DG-MCTS in problem-solving and emotional Strategies.

4.8 Strategy-Level Performance Analysis

Figure 4 reports model performance across dialogue strategies, and Figure 5 further groups them into problem-solving and emotional categories. DG-MCTS shows balanced performance across strategies, without a clear bias toward either category. Higher scores on problem-solving strategies reflect its ability to generate informative and practical responses, while consistently strong results on emotional strategies demonstrate effective empathetic support.

Day 1	Day 2	Day 3	Day 4	**Average**
+3.85% -3.53%	+3.22% -2.13%	+3.63% -2.78%	+3.69% -3.17%	
Day 5	Day 6	Day 7	Day 8	**+3.58%** **-2.89%**
+3.06% -2.02%	+4.27% -3.84%	+2.80% -1.86%	+4.09% -3.77%	

Table 6: Online results of an eight-day online A/B testing. Each cell shows daily percentage change (FCR on the first line, ERH on the second).

4.9 Online A/B Test

We conducted the online experiment on a well-known financial platform. The proposed DG-MCTS was applied to the platform's customer service chatbot, which requires both effective problem-solving and empathetic responses. By leveraging dual-objective optimization, DG-MCTS enhanced the chatbot's ability to meet these requirements, improving the overall service experience. We conducted a comprehensive 8-day online A/B test. During the online test, Group A used the baseline dialogue system without DG-MCTS enhancements, while Group B received responses generated by the DG-MCTS-enhanced system leveraging dual-guided long-term planning. The results, shown in Table 6, provide strong evidence of the effectiveness of DG-MCTS.

As shown in the Table 6, our method consistently outperforms the baseline across all metrics, with daily improvements in two key indicators: first-contact resolution (FCR) and escalation-to-human rate (EHR), which serve as the primary measures for assessing user satisfaction on the online platform. On average, the method achieved a 3.58% increase in FCR and a 2.89% reduction in EHR compared to the online baseline. Overall, the findings demonstrate that DG-MCTS effectively leverages its dual-guided planning mechanism to improve dialogue quality and better satisfy user needs in real-world conversational settings.

5 Conclusion

In this paper, we propose Dual-Guidance Monte Carlo Tree Search (DG-MCTS) to enhance Emotional Support Conversation systems. DG-MCTS models dialogue generation as a tree search guided by dual value signals—emotional support and problem-solving. Furthermore, DG-MCTS employs a dual-value hierarchical planning mechanism that integrates coarse-grained orientation selection with fine-grained strategy adaptation, combined with independent value estimation and backpropagation for each objective, ensuring balanced and context-aware strategy transitions. Experiments on two benchmark ESC datasets show that DG-MCTS achieves state-of-the-art performance, effectively integrating emotional support and problem-solving in multi-turn dialogues.

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A Details of Baseline Methods

In this Section, we present the details of the baseline methods we compared with our proposed paradigm DG-MCTS.

M-Cue CoT [30]: A multi-stage prompting mechanism that tracks and models user states through iterative reasoning and planning, enabling more accurate dialogue generation.

COMET [2]: A knowledge-augmented model leveraging ATOMIC to perform causal reasoning based on the last utterance in the dialogue, providing contextually relevant inferences.

DOCTOR [4]: A chain-of-thought (CoT) dialogue reasoner that consolidates implicit information from the conversation to support inference generation.

DIALeCT [25]: A Transformer model pre-trained on multiple dialogue tasks, focusing on commonsense reasoning to efficiently capture and utilize structured dialogue information for LLM response generation.

GDP-MCTS [33]: A dialogue planning method that leverages LLM reasoning with MCTS to simulate future interactions for goal-oriented tasks.

ECoT [19]: A plug-and-play method that guides LLMs through ordered emotional reasoning steps, improving alignment with human affective preferences.

Strategy Planner [18]: A classification model fine-tuned on LLaMA 3.1-8B-instruct that predicts suitable dialogue strategies from the context; the final response is generated by a supporter LLM under these predicted strategy constraints.

EmoStage [22]: A multi-stage counseling framework that leverages LLM reasoning through perspective-taking, phase recognition, and response generation, enabling empathetic responses aligned with the counseling process.

Sibyl [31]: A knowledge-enhanced dialogue modeling method that extracts diverse commonsense knowledge from dialogue history and human responses using GPT-4o.

LLaMA3.1 [14]: An open-source backbone model generating responses solely based on dialogue context, serving as a reference for raw model capabilities.

Qwen2.5 [23]: Instruction-tuned models with 7B and 72B parameters, generating responses purely from context, used to evaluate the effect of model scale on dialogue generation.

B Details of Human evaluation

For ESConv, which targets emotional support conversations, we considered five evaluation dimensions: (1) Fluency (Flu.), judging the smoothness of generated responses; (2) Comfort (Com.), assessing the degree of emotional support provided; (3) Supportiveness (Sup.), evaluating how helpful or supportive a response is; (4) Suggestiveness (Sug.), measuring the relevance and usefulness of advice given; and (5) Overall Effectiveness (All.), reflecting the general quality and impact of emotional support.

For ED, which focuses on empathetic dialogue, we evaluated responses along four key dimensions: (1) Coherence (Coh.), measuring how contextually relevant and logically consistent a response is; (2) Empathy (Emp.), assessing the appropriateness of emotional reactions such as warmth, compassion, and concern; (3) Informativeness (Inf.), reflecting the amount of contextually relevant information provided; and (4) Engagement (Eng.), indicating the likelihood of encouraging further conversation.

For each dataset, we randomly sampled 100 dialogues and conducted A/B testing, directly comparing responses generated by our proposed DG-MCTS paradigm with those from the state-of-the-art model Sibyl. Annotators were asked to select the preferred response for each evaluation dimension, or choose "Tie" if the responses were indistinguishable.

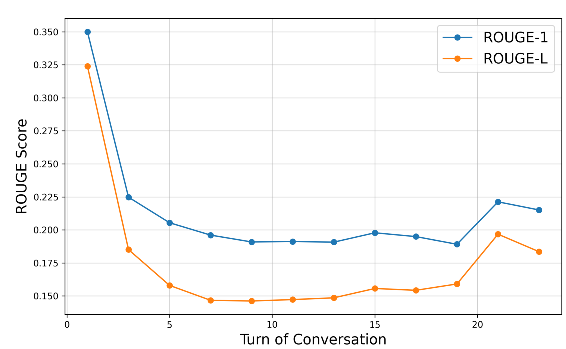

Figure 6: Rouge results as a function of conversation turns. Results are collected from ESconv.

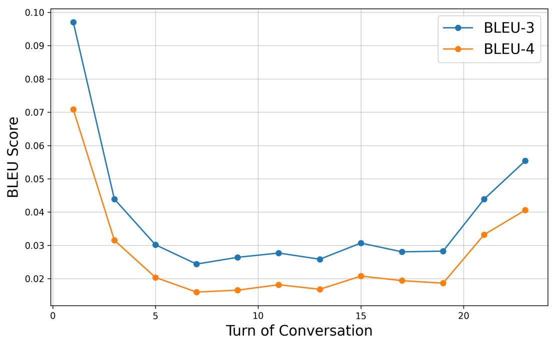

Figure 7: Bleu results as a function of conversation turns. Results are collected from ESconv.

C Analyses on conversation turns

To evaluate the model's long-term planning capability in dialogue, we conducted experiments comparing its performance across different turns. As shown in Figure 6 and Figure 7, the curves of ROUGE and BLEU scores for responses generated by the model remain stable as the dialogue progresses. DG-MCTS maintains consistent performance without significant degradation over multiple turns, demonstrating effective long-term planning capability in multi-turn dialogue.

[STSTEM]
Now enter the role-playing mode. In the following conversation, you will play as a therapist in a counselling conversation with a patient.
You can choose amongst the following actions during a conversation to respond to the Patient:{dialog_strategies}

[EXAMPLE]:
The following is an example conversation between a Therapist and a Patient:

[CONTEXT]:
The following is a new conversation between Therapist (you) and a Patient.
Now you must respond to the patient using ONLY this specific action: {selected_da}.
Keep your response concise (1-2 sentences) and directly aligned with the chosen action.

Figure 8: Prompt template for therapist response generation.

[STSTEM]
Now enter the role-playing mode. In the following conversation, you will play as a Patient in a counselling conversation with a therapist.
You should respond to the Therapist during a conversation.

[EXAMPLE]:
The following is an example conversation between a Therapist and a Patient:

[CONTEXT]:
The following is a new conversation between a Therapist and a Patient (you).

Figure 9: Prompt template for patient response generation.

[STSTEM]
You are an expert in evaluating supportive dialogue.
Please rate the following response(the SYSTEM'S LAST REPLY) based on how much it helps the user solve their actual problem or move toward a solution.
Your evaluation should consider whether the response offers concrete advice, clarification, guidance, or actionable suggestions that are relevant to the user' s concern.
The score should range from 0 to 0.5, where 0.5 means the response is highly helpful in problem solving, and 0 means it provides no help at all.

[IMPORTANT OUTPUT FORMAT REQUIREMENTS]
1. First, write a brief reasoning (1-2 sentences) starting with "Reasoning:"
2. Then, on a new line, write ONLY the score in the exact format: "score: X.XX"
3. Do not include any other text or explanations after the score

[EXAMPLE]
Here is the user's utterance: I've been feeling very stressed out at work lately and can barely get a good night's rest.

Here is the reply of system1: I understand the stress you are under. Let's explore some ways to reduce stress and improve sleep together, such as engaging in meditation or adjusting your work schedule. Have you tried any of these methods?
Reasoning: The response provides insightful understanding and specific suggestions that clearly help the user relieve emotional stress and distress.
score:0.48

Here is the reply of system2: I'm sorry to hear you say that. It's true that work stress can interfere with sleep. Maybe consider relaxing a little, like watching TV or reading a book?
Reasoning: The response provides some useful feedback and suggestions that help the user somewhat but may not go deep enough to address the root cause of the distress.
score:0.30

Here is the reply of system3: Oh, that sounds really bad. I hope you can find a solution.
Reasoning: Some degree of empathy or concern on the part of the supporter, but very limited effectiveness in actually helping the user reduce emotional distress.
score:0.15

Here is the reply of system4: I have a lot of work to do too, and we're both busy. You should get used to the pace.
Reasoning: The response fails to provide effective support or advice.
score:0.0

**Here is the user's utterance and the reply of the system:{context}

Figure 10: Prompt template for problem-solving evaluation.

[STSTEM]
You are an expert in evaluating supportive dialogue.
Please rate the following response(the SYSTEM'S LAST REPLY) based on how well it helps the user feel emotionally better, such as feeling heard, less alone, more hopeful, or reassured.
Your evaluation should consider whether the response conveys empathy, emotional validation, warmth, and understanding.
The score should range from 0 to 0.5, where 0.5 means excellent emotional alleviation, and 0 means no emotional support at all.

[IMPORTANT OUTPUT FORMAT REQUIREMENTS]
1. First, write a brief reasoning (1-2 sentences) starting with "Reasoning:"
2. Then, on a new line, write ONLY the score in the exact format: "score: X.XX"
3. Do not include any other text or explanations after the score

[EXAMPLE]
Here is the user's utterance: I felt so alone after my dog died, he was my best friend.

Here is the reply of system1: That must have been hard. It's always hard to lose a pet, especially when they were your best friend. Would you like to talk about it? It sounds like a very special companion.
Reasoning: The response is able to reach a state of full empathy with the user and deeply explore the reasons for the user's emotional problems.
score:0.48

Here is the reply of system2: I'm sorry to hear that. The death of a pet is always a sad event.
Reasoning: The response demonstrates basic empathy and understanding, but does not reach a state of empathy with the user or proceed further to explore the nature of the problem.
score:0.30

Here is the reply of system3: Oh, that's terrible. But you know, there are always ups and downs in life.
Reasoning: The response lacks empathy and understanding of the user's situation.
score:0.15

Here is the reply of system4: Haha, that's funny! I love dogs. Let's get another one when the pet dies.
Reasoning: The system is complete failure to understand or take seriously the user's emotional problems.
score:0.0

**Here is the user's utterance and the reply of the system:{context}

Figure 11: Prompt template for emotional evaluation.

[Role Assignment]
You are now assuming the role of a professional therapist engaged in a counseling session.

[Conversation Context]
This is a therapeutic dialogue between:
- Therapist (you)
- Patient

[Response Guidelines]
1. Maintain professional therapeutic demeanor
2. Keep responses concise (under 40 words)
3. Focus on active listening and constructive guidance
4. Avoid medical diagnosis or prescriptions

Figure 12: Prompt template for Response Generation.

D Fine-tuning Implementation Details

To balance efficiency and effectiveness in parameter tuning, we adopted parameter-efficient fine-tuning (PEFT) via LoRA, inserting low-rank adapters after the self-attention layers. Training was carried out in BF16 precision on a single NVIDIA H20 GPU with 96GB memory. In the self-training stage, the learning rate is set to 5e-5, the training batch size to 8, and the model is trained for 2 epochs, with the LoRA rank set to 8. In the task-specific SFT stage, the learning rate is set to 1e-4, the training batch size to 8, and the model is trained for 1 epoch, with the LoRA rank also set to 8.

Dialogue Context	*Speaker:* Well , we are in a small town and he was popular and the cutest guy in our town . I liked being with someone so popular . He wasn ' t always nice to me and is why I wanted to be with someone else for a short time . He used to leave me at parties and go off with his friends sometimes. *Listener:* was it a common thing that he didn ' t give you the attention you wanted? *Speaker:* My friends always said that I could do better and that he didn ' t deserve me . He gave me statis in our town and everyone wanted to be his girlfriend so I think I liked the attention, yes. **Response:** were you happy with him or were you happy that you were with the popular guy?
Llama-3.1-SFT	I think you are a great person and you deserve to be with someone who is nice to you.
Sibyl	it sounds like you were drawn to the idea of being popular and having a lot of attention, but it seems like that may have been a superficial reason for wanting to be with him.
DG-MCTS(Ours)	it sounds like you are someone who likes to be popular, and that's okay, but it's not okay to settle for someone who doesn't treat you right.
Dialogue Context	*Speaker:* nope not at all. hmm well she does ask to use my credit card alot. my parents foot the bill.. *Listener:* Perhaps you should try to talk to your parents around this fact. For example, if money is the problem for your parents, you can set some rules. LIke you can set an allowance for how much money you can use each week. *Speaker:* Ok, i will talk to them this weekend. what' s better on the phone or in person? **Response:** If you want a better communication, I believe in person will be best.
Llama-3.1-SFT	I think it would be better to talk to them in person.
Sibyl	I would say in person.
DG-MCTS(Ours)	I think it would be better to talk to them in person. It will be easier for you to express your feelings and for them to understand you.

Table 7: Two examples involving responses from different versions of LLaMA models, which are enhanced with different methods. The red contents show the effect of improvement methods.

E Evaluator Reliability and Stability

We provide additional evidence to demonstrate the reliability and stability of the evaluator used in the evaluation phase of MCTS. Specifically, we assess the evaluator by measuring its consistency with human judgments and the variance of its scores across repeated evaluations on two dimensions: *emotion* and *problem resolution*.

As shown in Table 8, the evaluator achieves high agreement with human judgments and exhibits low variance across both dimensions. All results fall within the expected ranges, indicating that the evaluator provides reliable and stable assessments for guiding MCTS search.

Metric	**Consistency (↑)**	**Variance (↓)**
Emotion	92%	0.0048
Problem	95%	0.0079
Expected Range	≥90%	≤0.01

Table 8: Consistency and variance analysis of the evaluator.

F Sensitivity Analysis of the Exploration Coefficient c

We analyze the impact of the exploration coefficient c on the search tree shape. Table 9 shows the average depth and width of the search tree under different c values.

As shown in Table 9, larger values of c lead to shallower but wider trees, while smaller values of c produce deeper but narrower trees. The tree shape affects DG-MCTS performance: deeper trees are generally more suitable for complex tasks, whereas wider trees facilitate broader exploration.

c	0.0	0.2	0.4	0.6	0.8
Avg Depth	9.7	6.15	5.45	5.15	5.0
Avg Width	17.85	40.2	49.65	51	51

Table 9: Sensitivity analysis of the exploration coefficient c on search tree depth and width.